W-A-450/12
JANUARY 3, 1992

FEDERAL SPECIFICATION

Detector, Motion, Video, Interior (Closed Circuit Television)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type XII, Closed Circuit Television Motion Detector/Interior Video Motion Detector (IVMD).

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposals shall apply.

Electronic Industrial Association (EIA) Publications:

EIA Standard RS170 Composite Video

3. REQUIREMENTS.

3.1 Qualification. The IVMD furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been

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tested and shall have passed the qualification tests specified in 4.1.

3.2. Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

All units shall withstand the tests in 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 20 000 hours.

3.4. Operation. The IVMD shall be capable of designating an adjustable portion of a video camera scene as an alarm area and detecting an intruder by motion sensing. The detector shall, after initial setup, stabilize within five minutes after being powered on and after any alarm initiating stimulus is removed. The unit shall be capable of receiving the input of one video camera and providing the alarmed scene to an external video monitor or signalling an external video switcher to lock on to the alarmed camera. If the IVMD is capable of monitoring more than one camera, then an intrusion should automatically switch the alarmed scene to the output.

Positioning/adjustment of the alarm zones shall be portrayed by either a visible outline or a visible highlight over the video scene. This outline/highlight shall be capable of being switched off while the alarm function is retained.

Interior video motion detectors are exempt from the "power failure" requirements of 3.5 of W-A-450C/GEN.

3.4.1 Detection parameters. The detector shall be capable of detecting a person comprising at least 10% of the vertical height and 10% of the horizontal width of the video scene, moving within the detection zone at a rate of 0.3 m/s or faster. The

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sensitivity of the alarm zone (reaction to contrast change) shall be adjustable to provide compromise between probability of detection and nuisance alarms.

3.4.2 Inputs/outputs.

3.4.2.1 Power. A.C. power input shall be a three conductor (AWG 18) stranded copper cable at least five feet in length and terminated with a U-Ground Male Plug.

3.4.2.2 Video. Camera inputs and monitor outputs shall be on a chassis mounted female BNC coaxial connectors. Signal requirements are EIA Standard RS-170 Composite Video (1 volt, peak to peak) for both camera inputs and monitor outputs.

3.4.2.3 Alarm outputs. The IVMD shall provide passive alarm contacts to activate an external alarm system. These contacts shall be externally accessible, either on individual screw terminals, or a chassis connector for which a mating connector is provided. This connector must be capable of carrying 250 milliamperes of current at 48 volts.

3.4.3. Alarms. The IVMD shall provide both a visible and audible local alarm. These local alarms shall be capable of being reset manually by the operator, or automatically by a timer.

3.4.4 Alternate processing format. IVMD units listed by the manufacturer as capable of operating on 220V, 50Hz, must be compatible with Closed Circuit Infrared (CCIR) Video format of 625 scan lines, 50Hz vertical sync pulses.

3.4.5 Statement of supplied/optional capability. While the following options are not mandatory, the IVMD supplier shall state whether or not the following capabilities are either supplied or available on their unit:

(a) Both Normally Open (N/O) and Normally Closed (N/C) contacts on the external alarm circuit.
(b) Loss-of-video alarm (indication of loss of video signal amplitude while retaining horizontal and vertical sync pulses).
(c) Loss-of-sync alarm (indication of total video and sync loss)

3.4.6 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature concerning performance data that would exceed the minimum requirements of this specification.

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3.5 IVMD Enclosure. All equipment supplied under this specification shall be in an enclosure suitable to be co-located with other interior security control equipment. The enclosure shall protect the operating controls from physical damage and provide total electrical protection to the operator. The IVMD units are exempt from the "Tamper/Access" requirements of 3.3 and 3.5 of W-A-450C/GEN.

3.6 Switch and/or relay assembly. Alarm switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the published current rating shall not cause damage to the switch or contacts.

3.7 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.8 Maintainability. The IVMD shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover.

3.9 Life cycle. The IVMD alarm switches and/or relays shall be rated at not fewer than 1 000 000 operations under a load of 250 milliampere driven from a 48 volt DC source. Contact resistance shall not change more than +20% from the initial reading.

3.10 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests.

4.1.1 Performance test. The unit being tested shall be mounted, wired, and aligned in accordance with the manufacturer's recommendations. The unit's alarm contacts shall be connected to

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a buzzer. The unit shall be tested to determine conformance with 3.4.1.

4.1.2 Stability test. The equipment submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.10 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference

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in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil Agency Marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

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6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.2 Invalid alarm. Any other alarm.

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